Drug-Safety Intelligence Platform

Project Documentation

Project Planning & Management

Literature Review

Requirements Gathering

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Graduation Project

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1. Project Planning & Management

This section defines what the project is, what it will and will not deliver, and how the work is organized, scheduled, and monitored.

1.1 Project Overview

The Drug-Safety Intelligence Platform is a data engineering project that collects, standardizes, and analyzes drug adverse-event reports from the FDA Adverse Event Reporting System (FAERS), obtained from the FAERS quarterly data files or through the openFDA Drug Adverse Event API (the final ingestion route is a design decision, see Section 2.2.7). The platform integrates this data with standardized drug information from RxNorm to resolve inconsistent drug names, map reports to the correct medications, and reduce duplicate records.

After ingestion and transformation, the platform builds an analytical data model that enables users to explore adverse-event patterns by drug, active ingredient, reaction, demographics, reporter type, outcome severity, and time period.

The platform aims to transform fragmented and inconsistent drug-safety data into a clean, standardized, and analysis-ready dataset that can support drug-safety monitoring and exploratory analysis.

1.2 Problem Statement

Drug-safety data is available through public sources such as FDA FAERS, but the raw data can be difficult to analyze directly due to its size, inconsistent drug naming (different brand or generic names), duplicate reports, incomplete records, and different representations of the same medication.

Furthermore, raw adverse-event data contains multiple related entities such as patients, drugs, reactions, outcomes, and report information, requiring proper data integration and transformation before meaningful analysis can be performed. The FDA's own documentation notes the presence of duplicate and incomplete reports in the AEMS/FAERS data (see Section 2.2.3).

In summary, the platform must:

- Integrate data from different sources.
- Clean and validate raw records.
- Standardize drug identities.
- Detect and remove duplicate records.
- Build relationships between drugs and adverse reactions.
- Produce reliable analytical datasets.
- Present the results through an interactive interface.

1.3 Project Objectives

Primary Objective

To design and implement an end-to-end data engineering pipeline that transforms raw drug-safety data into a clean, standardized, and analytics-ready platform for exploring adverse-event patterns.

Specific Objectives

O1. Data Ingestion

- Collect adverse-event data from FDA FAERS/openFDA.
- Collect standardized drug concepts and identifiers from RxNorm.

O2. Data Cleaning

- Handle missing and invalid values.
- Normalize text fields.
- Standardize formats and data types.

O3. Drug Standardization

- Map different drug-name representations to standardized drug concepts.
- Assign consistent identifiers to medications using RxNorm.

O4. Deduplication

- Identify and handle duplicate or repeated safety reports.
- Prevent duplicate records from distorting analytical results.

O5. Data Integration

- Integrate FDA adverse-event information with standardized drug information.
- Establish relationships between drugs, adverse reactions, outcomes, and reports.

O6. Data Modeling

- Build an analytical data model optimized for querying and reporting.

O7. Safety Analytics

- Analyze the frequency of adverse reactions.
- Identify common drug–reaction relationships.
- Analyze trends over time.
- Compare adverse-event patterns across drugs or drug classes.

O8. Visualization

- Develop an interactive dashboard for exploring the processed data.

O9. Data Quality & Reliability

- Implement validation and data-quality checks throughout the pipeline.

1.4 Project Scope

Data Sources

- FDA FAERS / openFDA adverse-event data.
- RxNorm standardized medication data.

Data Engineering

- Data extraction and ingestion.
- Data transformation.
- Data cleaning.
- Missing-value handling.
- Drug-name normalization.
- Drug entity resolution.
- Duplicate detection.
- Data validation.
- Data integration.

Data Storage

- Structured storage of raw and processed data.
- Analytical data warehouse / database.
- Proper relational or dimensional data modeling.

Analytics

- Adverse-event frequency analysis.
- Drug–reaction relationship analysis.
- Temporal trend analysis.
- Drug/class comparison.
- Severity/outcome analysis where available.

Visualization

- Interactive dashboard.
- Drug search.
- Reaction search.
- Top adverse reactions.
- Drug–reaction relationships.
- Trends over time.
- Filters by drug, reaction, date, and other available attributes.

Out of Scope

To keep the project technically realistic and avoid overclaiming its medical capabilities:

- The platform will not provide medical diagnosis.
- It will not recommend or prescribe medications.
- It will not establish causal relationships between drugs and adverse events.
- It will not replace professional pharmacovigilance or clinical decision-making.
- Drug–drug interaction checking is not included, because commercial interaction data sources are restricted (see Section 3.6).
- Predictive machine-learning models for clinical risk prediction are outside the initial project scope.

1.5 Project Timeline

Table 1.1 presents the main phases of the project, from planning to final documentation, with the start and end date and the member responsible for each phase. The project runs from 24 October to 1 December 2026 (39 calendar days, about 5.5 weeks), and some phases overlap to fit the short schedule. Durations are counted in calendar days, including the start and end dates. Figure 1.1 shows the same plan as a Gantt chart, so the order and overlap of the phases can be seen at a glance.

Table 1.1: Project phases and timeline

Phase	Dates (2026)	Days	Owner
1. Planning Scope, objectives, plan	24 Oct – 27 Oct	4	Randa Gomaa, Mariam Abdelaziz
2. Literature review FAERS, RxNorm, similar systems	26 Oct – 2 Nov	8	Youssef Asaad Saad, Olivia Hanna
3. Requirements Functional, technical, criteria	30 Oct – 5 Nov	7	Abanoub Ayman
4. Data collection Download, clean, map to RxNorm	31 Oct – 12 Nov	13	Data Engineer
5. System design Architecture, database, UI	3 Nov – 10 Nov	8	Backend, Frontend
6. Development Pipeline, signals, UI	10 Nov – 23 Nov	14	Dev team
7. Testing Unit, integration, user tests	22 Nov – 27 Nov	6	QA
8. Documentation Merge, PDF, GitHub	24 Nov – 1 Dec	8	All members

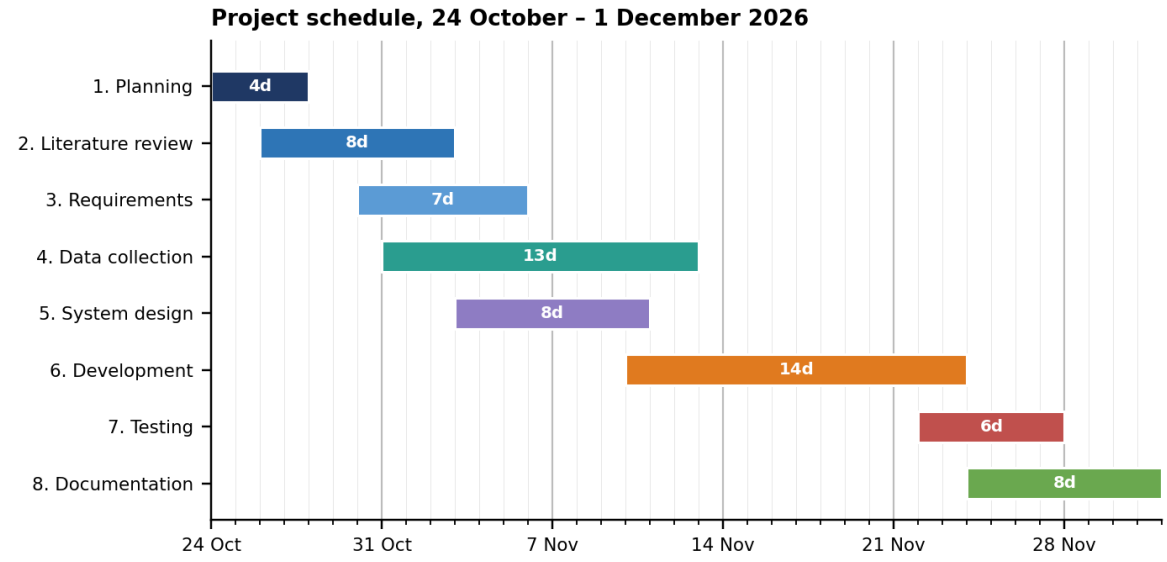


Figure 1.1: Project Gantt chart.

1.6 Project Milestones

Table 1.2 lists the eight project milestones with their target dates and expected deliverables. These milestones are used to track progress and to measure on-time delivery.

Table 1.2: Project milestones

Milestone	Date (2026)	Deliverable
M1. Project kickoff	27 Oct	Approved scope and plan
M2. Literature review done	2 Nov	Review section and references
M3. Requirements approved	5 Nov	Requirements specification
M4. Design approved	10 Nov	Architecture and UI mockups
M5. Clean dataset ready	12 Nov	FAERS cleaned, mapped to RxNorm
M6. First prototype	17 Nov	Working prototype (search, signals)
M7. Testing complete	27 Nov	Test report and fixes
M8. Final documentation	1 Dec	Final PDF uploaded to GitHub

1.7 Team Roles and Responsibilities

Table 1.3 lists the team members and the documentation work that each member has completed so far. Youssef Asaad Saad is the team leader.

Table 1.3: Team roles and responsibilities

Team member and role	Work completed so far	Sections
Randa Gomaa Team Member	Project overview, problem statement, objectives, and scope	Sections 1.1 to 1.4
Mariam Abdelaziz Team Member	Project planning and management: timeline, milestones, team roles, risks, and success metrics	Sections 1.5 to 1.9
Youssef Asaad Saad Team Leader	Literature review: introduction, data sources (FAERS and RxNorm), and similar projects and systems	Sections 2.1 to 2.3
Olivia Hanna Team Member	Comparison with existing solutions, limitations and research gap, and stakeholders	Sections 2.4, 2.5, and 3.1
Abanoub Ayman Team Member	Requirements: functional, non-functional, data, technical, constraints, and acceptance criteria	Sections 3.2 to 3.7

Note: the technical responsibilities (data engineering, backend, frontend, and testing) have not been assigned yet and will be added here once they are agreed.

1.8 Risk Management

Table 1.4 identifies the main risks that may affect the project, rates each one by probability and impact, and proposes a mitigation action. The highest risks relate to data quality, inconsistent drug names, and the limited project duration.

Table 1.4: Risks and solutions

Risk	Prob. / Impact	Mitigation
R1. FAERS data is large, with duplicates and errors	High / High	Clean early, remove duplicates, work on a sample first
R2. Drug names are not standardized	High / Medium	Map names to RxNorm and review unmatched ones manually
R3. Short timeline (about 5 weeks)	High / High	Limit scope, overlap phases, follow the milestones in Table 1.2
R4. A team member is delayed or unavailable	Medium / High	Weekly check-ins, peer reviewers, backup owner per task
R5. Slow queries on large data	Medium / Medium	Add database indexes, use a data subset, cache results
R6. Inaccurate signal detection results	Medium / High	Use standard methods, validate against known drug signals
R7. Merge conflicts and inconsistent formatting	Medium / Low	Shared Word styles, Git branches, one final editor
R8. Data privacy concerns	Low / High	Use only public, anonymized data and add a disclaimer

1.9 Success Metrics

Table 1.5 defines measurable targets used to judge whether the project has succeeded, such as the RxNorm mapping rate, search response time, signal detection accuracy, and on-time delivery. Each metric has a target value and a method of measurement.

Table 1.5: Success metrics and target values

Metric	Target	How it is measured
S1. RxNorm mapping rate	90% or more	Mapped drug names / total drug names
S2. Duplicate removal	95% or more	Duplicates removed in a sample check
S3. Search response time	Under 3 seconds	Average time over 20 test queries
S4. Signal detection accuracy	80% or more	Match with known drug-safety signals
S5. Requirements implemented	90% or more	Requirements that pass acceptance criteria
S6. Test pass rate	90% or more	Passed test cases / total test cases
S7. On-time delivery	8 of 8 milestones	Milestones met by the dates in Table 1.2
S8. User satisfaction	4 out of 5	Short survey after user testing

2. Literature Review

2.1 Introduction

Pharmacovigilance concerns the monitoring of the safety of medicines after they have been approved and placed on the market, a task often described as post-marketing safety surveillance [1]. Identifying adverse drug reactions during this post-marketing phase is regarded as one of the most important goals of drug safety surveillance, and spontaneous reporting systems, which remain the mainstay of traditional surveillance, are used mainly to generate hypotheses rather than to confirm them [2].

Large reporting systems exist at both national and international levels. In the United States, the FDA Adverse Event Reporting System (FAERS) is designed to support the Food and Drug Administration's (FDA) post-marketing safety surveillance program for drug and therapeutic biologic products [1]. Internationally, the Uppsala Monitoring Centre (UMC) maintains VigiBase, a worldwide database of spontaneous individual case safety reports (ICSRs) that operates under the World Health Organization (WHO) Programme for International Drug Monitoring [3].

The usefulness of these systems depends on how the reports are analyzed. Statistical screening methods have been developed to help reviewers find potential signals in large volumes of spontaneous reports. Evans et al. likened the search for previously unrecognized hazards in such data to looking for a needle in a haystack, and proposed the proportional reporting ratio (PRR) as a simple statistical aid to signal generation [4]. Bate and Evans later reviewed the main quantitative methods in use, including the PRR, the reporting odds ratio, the information component, and the empirical Bayes geometric mean [5]. Results obtained in this way are treated as hypotheses. In the WHO vaccine-safety setting, for example, signals detected by a screening algorithm are reviewed clinically by pharmacovigilance experts, and signals are regarded as hypotheses that must be verified and further evaluated [3].

Raw reports are rarely ready for analysis. Banda et al. report that the publicly available FAERS data require substantial curation before they can be used appropriately, and that the data-cleaning and normalization strategies applied can have a material impact on analytical results [2]. Problems that must be addressed include duplicate case records and drug names that must be mapped to a standard vocabulary [2]. These are data engineering problems. They call for reliable ingestion of source files, explicit data-quality rules, terminology standardization, integration of heterogeneous sources, and transformations that can be repeated and audited, before any analytical or dashboard layer can be trusted.

The Drug-Safety Intelligence Platform proposed in this project is intended to combine FAERS adverse event reports with standardized drug information from RxNorm, and to present patterns in the reported events through an interactive dashboard. The emphasis is on data ingestion, quality, transformation, integration, and analytics. The platform is designed to analyze reported adverse events. It is not intended to prove that a particular drug caused a particular event, and it does not provide medical advice. The remainder of this section describes the two data sources (Section 2.2), reviews related systems and research projects (Section 2.3), compares them with the proposed platform (Section 2.4), and identifies the research gap (Section 2.5).

2.2 Data Sources: FDA FAERS and RxNorm

2.2.1 Overview and Purpose of FAERS

The FDA Adverse Event Reporting System (FAERS) is a database that contains adverse event reports, medication error reports, and product quality complaints that resulted in adverse events and were submitted to the FDA. It is designed to support the FDA's post-marketing safety surveillance program for drug and therapeutic biologic products. Its informatic structure follows the international safety reporting guidance of the International Conference on Harmonisation (ICH E2B), and adverse events and medication errors are coded using the Medical Dictionary for Regulatory Activities (MedDRA) [1].

Terminology is currently changing. The FDA's FAERS web page is now titled "FDA Adverse Event Monitoring System (AEMS)" and carries the note "formerly FDA Adverse Event Reporting System (FAERS)". The FDA states that it is implementing AEMS to consolidate multiple reporting systems used across FDA-regulated product categories. The same page continues to describe the FAERS database and to link to the FAERS Public Dashboard and the quarterly data files [1]. This document therefore uses the name FAERS for the drug and therapeutic biologic adverse event database, which is the name used in most of the literature and in the data files, and notes that the FDA is consolidating it under AEMS.

FAERS data are made available in several forms. The FDA offers an interactive FAERS Public Dashboard and continues to provide the FAERS quarterly data files [6]. openFDA, a separate FDA service, publishes FAERS data with quarterly updates and a lag of about three months [7].

2.2.2 Collection and Structure of Reports

FAERS holds spontaneous reports. When a manufacturer receives an adverse event report, it is normally required to send the report to the FDA [8]. Reports may also be sent to the FDA by other parties, such as health professionals and consumers [6]. FAERS replaced the earlier legacy AERS system on September 10, 2012 [9], and the FDA's documentation describes the current database as case- and version-based, in contrast with the report-based identifier (the ISR) used previously [9]. Reports are not edited in place: subsequent updates are submitted as separate reports [8]. As a result, one clinical case can be represented by more than one record.

The public quarterly data are distributed as relational extracts. The FDA notes that a simple search cannot be performed on these files by people who are not familiar with building relational databases [6]. Within each quarterly package, the information about a case is split across separate related files, for example demographic, drug, reaction, outcome, report-source, therapy-date, and indication data, which must be joined using report identifiers. openFDA offers a different representation: it converts the data into JSON and annotates the original records with additional fields [8]. These two access routes therefore deliver the same underlying reports in structurally different forms, which matters when designing an ingestion pipeline.

2.2.3 Limitations and Data Quality Challenges of FAERS

FAERS is a valuable source of signals, but its limitations are stated explicitly by the FDA. Reports do not need to establish that a product caused an event, and reports do not always contain enough detail to evaluate the event properly; consequently, FAERS data cannot be used to calculate the incidence of an adverse event in the United States [6]. The FDA also notes that its public data contain duplicate reports and that some reports do not contain all the necessary information [6].

openFDA adds that adverse event reports undergo no extensive validation or verification, that a causal relationship therefore cannot be established between a product and the reactions listed, and that the FDA does not receive reports for every adverse event or medication error that occurs with a product [8]. The FDA's readme also cautions that a reported event may have been related to the underlying disease being treated or caused by another drug [9].

Reporting counts are further affected by usage. Many drug products appear frequently in reports simply because many people take them, not because they cause more adverse events [8]. Raw report counts therefore cannot be compared across drugs as if they measured risk.

From a data quality perspective, duplicate case records are a recognized problem: the curated FAERS resource of Banda et al. explicitly removes duplicate case records and applies standardized vocabularies, because the source data require substantial curation [2]. Duplication is not unique to FAERS, as the UMC also notes that VigiBase can contain duplicate reports [10]. Drug names in the source data must also be standardized before drug-level analysis is reliable [2].

2.2.4 Overview of RxNorm and Its Identifiers

RxNorm is a standardized nomenclature for clinical drugs produced by the U.S. National Library of Medicine (NLM) [11]. It provides normalized names and unique identifiers for medicines, and it contains the names of prescription drugs and many over-the-counter drugs available in the United States [12]. An RxNorm clinical drug name combines the ingredients, strengths, and dose form of a drug [11], and RxNorm also creates names at other levels of specificity [12]. RxNorm's standard names are connected to the varying names used in many other drug vocabularies, with the aim of supporting interoperability among computer systems that record or process drug data [11].

The central identifier is the RxNorm Concept Unique Identifier (RXCUI). Every name and code supplied to NLM by a source vocabulary receives an RXCUI [12]. NLM's own tutorial illustrates the idea with naproxen: differently written source names have the same meaning at a certain concept level, and RxNorm groups these synonyms into one normalized concept, naproxen 250 MG Oral Tablet, with the RXCUI 198013 [13]. RxNorm also maps the FDA National Drug Code (NDC), a 10- or 11-digit identifier assigned to medications in the United States [13]. A single RxNorm clinical drug concept may have several NDCs, whereas a single NDC is expected to be assigned to only one RXCUI [11].

RxNorm does not cover every string that a source supplies. NLM reports that about 60% of the drug names from RxNorm's source vocabularies receive RxNorm normalized names, while the remainder are out of scope or too ambiguous, with medical devices, foods, and enzymes the most common examples [12]. This figure describes RxNorm's own source vocabularies and must not be read as the mapping rate to be expected for FAERS drug names. RxNorm clinical drug names also do not differentiate between over-the-counter and prescription medications [11]. Mapping to RxNorm should therefore be treated as a process whose success must be measured, and not as one that is guaranteed to be complete or one-to-one.

Mapping to RxNorm has precedent in the FAERS literature. Banda et al. mapped FAERS drug names to RxNorm concepts as part of their standardization pipeline [2], and the authors' description of that work states that drug names were standardized to RxNorm ingredients, and, for multi-ingredient drugs, to standard clinical names [14].

2.2.5 Complementary Roles and Integration Challenges

FAERS and RxNorm answer different questions. FAERS records what was reported: the patient, the drugs named in the report, the events, and the outcomes. RxNorm provides a standardized reference for what a drug name denotes. Linking the two allows drug-level aggregation that does not depend on how a reporter happened to write a name, which is the objective that motivated the curated FAERS resource of Banda et al. [2]. Table 2.1 summarizes the main integration challenges discussed in the literature and the way the project proposes to handle them. The right-hand column describes proposed design choices, not implemented functionality.

Table 2.1: Integration challenges and proposed handling

Challenge	What the literature and documentation indicate	Proposed handling (design choice)
Inconsistent drug names	Source vocabularies format drug names in many ways [13], and FAERS drug names must be standardized before drug-level analysis [2].	Normalize text (case, spacing, punctuation), then match to RxNorm names, with manual review of the most frequent unmatched strings.
Duplicate and multiple-version reports	FAERS is case- and version-based [9], updates arrive as separate reports [8], and curated resources remove duplicate case records [2].	Keep one version per case, and evaluate further duplicate-detection rules. Record how many records are removed and why.
Unmappable names	Not every name receives a normalized RxNorm name [12].	Retain unmapped records with an explicit status flag, and report the mapping coverage as a measured data quality metric.
Different data structures	FAERS quarterly extracts are relational files [6], openFDA provides annotated JSON [8], and RxNorm is organized around concepts and identifiers [11].	Load each source into staging tables, document the schemas and join keys, and transform to a common analytical model.
Granularity of drug concepts	RxNorm has names at several levels of specificity [12], and one clinical drug concept may have several NDCs [11].	Select the analytical grain, for example ingredient level (as in [14]), and document the choice.

2.2.6 Reported Associations Versus Causal Relationships

A FAERS record states that a reporter suspected a link between a product and an event. It does not demonstrate that the product caused the event. The FDA does not require a causal relationship to be proven for a report to be accepted [6], and openFDA states that the reports have not been scientifically or otherwise verified as to cause and effect [8]. The FDA's readme further notes that the placement of an event code alongside a suspect drug does not imply an association with that specific drug [9]. For these reasons the FDA states that FAERS data cannot be used to calculate the incidence of an adverse event [6].

Disproportionality statistics such as the PRR compare the reporting of an event with one drug against the reporting of the same event with other drugs in the database [4]. They are screening aids for signal generation [4], [5], and the resulting signals are hypotheses for further evaluation [3]. In this project, any such statistic will be described as a measure of reporting patterns and not as a measure of risk or causation. Dashboard outputs will use wording such as "reported" and "reporting", and will carry a clear notice that the results do not constitute medical advice.

2.2.7 Proposed Integration Approach

The following steps describe the integration approach that the project proposes. They represent the intended design, subject to refinement during the design and implementation phases, and should not be read as a description of completed functionality.

1. The platform will ingest FAERS data from the quarterly data files, or alternatively from the openFDA API, and will retain the raw data unchanged in a staging layer.
2. It will apply data-quality checks and cleaning rules, including standardization of text fields and validation of identifiers.
3. It will reduce duplication by handling multiple versions of the same case, and will evaluate additional duplicate-detection rules, recording the effect of each rule.
4. It will map reported drug names to RxNorm concepts, using RxNorm data obtained from NLM, and will keep unmapped names in the dataset with an explicit status so that mapping coverage can be measured and reported.
5. It will store the standardized data in an analytical model suitable for aggregation by drug concept and by event.
6. It will compute descriptive counts and exploratory reporting-pattern statistics and present them in an interactive dashboard, with the limitations described in Section 2.2.6 stated on the dashboard.

2.3 Similar Projects and Systems

This subsection describes existing systems and research projects that are relevant to pharmacovigilance data and to the analysis of reported adverse events. It is descriptive: the systems are presented in terms of their purpose, data sources, main capabilities, and relevance to this project. The detailed comparison with the proposed platform is presented separately in Section 2.4.

2.3.1 WHO VigiBase and VigiAccess

Purpose and data. VigiBase is the WHO global database of individual case safety reports. It is maintained by the Uppsala Monitoring Centre (UMC), the WHO Collaborating Centre for International Drug Monitoring [3]. Its reports are contributed by the national pharmacovigilance centres that belong to the WHO Programme for International Drug Monitoring [15]. The UMC guideline for using VigiBase in studies notes that reports originate from many different countries and sources, that the amount of information in each report varies, that VigiBase can contain duplicate reports, and that reports are continuously added, updated, and sometimes deleted [10].

Capabilities. The UMC screens VigiBase with a screening algorithm. Signals found in this way are reviewed clinically by pharmacovigilance experts before being communicated to the WHO programme [3]. VigiAccess is the public web interface to VigiBase data. The UMC and WHO state that VigiAccess cannot be used to infer a confirmed link between a suspected side effect and a medicine, to determine the likelihood of a side effect occurring, or to compare the safety profiles of different medicinal products [15].

Relevance. VigiBase is the international counterpart to FAERS and is built on the same kind of spontaneous ICSR data. Its documented problems with heterogeneity and duplication, and the explicit usage limits stated for VigiAccess, are consistent with the cautions applied to FAERS in this project [10], [15]. The project does not use VigiBase data.

2.3.2 openFDA

Purpose and data. openFDA is an FDA service whose drug adverse event API returns data collected through FAERS [8]. The reports follow the ICH E2b/M2 version 2.1 standard, openFDA converts the data into JSON and annotates the original records with additional fields, and the data are updated quarterly [8], [7].

Capabilities and relevance. openFDA gives programmatic access to FAERS records and repeats the FDA's warnings that the reports are not validated, that causality cannot be established, and that incidence cannot be estimated [8]. It is relevant to this project as a possible ingestion route for FAERS data, alongside the quarterly data files, and its JSON structure differs from that of the relational extracts [8], [6]. The choice between these routes remains a design decision for the project.

2.3.3 FDA FAERS Public Dashboard

The FDA also provides the FAERS Public Dashboard, an interactive web-based tool that lets the public query FAERS data [6]. The FDA states that it will continue to provide the FAERS quarterly data files, which it describes as intended for users who are able to build relational databases [6]. The dashboard is relevant as an official example of presenting FAERS data interactively, and its FAQ is a primary source for the limitations of the data discussed in Section 2.2.

2.3.4 OpenVigil FDA

Purpose and data. OpenVigil FDA is a web-based pharmacovigilance analysis tool that uses the openFDA online interface to access U.S. and international adverse event data from the FDA's adverse event reporting system [16]. It forms part of the wider OpenVigil project, whose other versions work on raw FAERS data (OpenVigil 1) and on data cleaned using DrugBank and Drugs@FDA (OpenVigil 2); the project itself does not produce or gather any of the pharmacovigilance data [17].

Capabilities. According to its authors, OpenVigil FDA provides disproportionality analyses in order to identify the drug most likely to be associated with a new adverse event, to compare two drugs, to check combinations of two drugs for unknown interactions, and to improve the relevance of results by identifying and removing confounding factors using background correction. It requires no special licenses or statistical programs [16]. The wider project lists support for the proportional reporting ratio and reporting odds ratio, export to spreadsheet and statistics programs, and a drug-name mapper that uses RxNorm [17].

Relevance. OpenVigil FDA is a published example of analytics built on FAERS data through openFDA. It shows that disproportionality analysis can be offered to users through a web interface, and that RxNorm has been used for drug-name mapping in this area [16], [17].

2.3.5 AEOLUS (Banda et al., 2016)

Purpose and data. Banda et al. provide a curated and standardized version of FAERS with the aim of reducing the time researchers spend on data management and of improving the quality of the data [2]. The resource is named AEOLUS (Adverse Event Open Learning through Universal Standardization) [2].

Capabilities. According to the authors, the resource removes duplicate case records, applies standardized vocabularies with drug names mapped to RxNorm concepts and outcomes mapped to

SNOMED CT concepts, and includes pre-computed summary statistics about drug–outcome relationships. The source code is released together with the data [2].

Relevance. AEOLUS is the closest published precedent for the data engineering core of this project: deduplication of FAERS case records and mapping of FAERS drug names to RxNorm [2]. It also demonstrates that cleaning and normalization choices influence analytical results [2]. Because it is a research resource published in 2016, its current maintenance status and the period of data it covers should be checked before any reliance on it.

2.4 Comparison with Existing Solutions

This section compares the proposed Drug-Safety Intelligence Platform with existing public systems that expose, analyze, or describe drug adverse-event data. The systems were chosen because each covers one part of the platform's scope: public access to FAERS (FDA Public Dashboard, openFDA), analysis tools built on FAERS (OpenVigil), a curated FAERS dataset (AEOLUS), and international spontaneous-report data (VigiBase and VigiAccess). They are the same systems described in Section 2.3; this section compares them with the proposed platform. RxNorm is not treated as a competitor, because the platform uses it as a reference terminology [12]. All information was taken from official documentation and peer-reviewed publications, plus one news report [18] for the date on which FDA moved the dashboard to daily updates; web sources were accessed on October 17, 2026. As explained in Section 2.2.1, FDA is consolidating FAERS under the Adverse Event Monitoring System (AEMS) [1]; this document keeps the name FAERS, which the data files and most of the literature still use.

2.4.1 Evaluation Criteria

1. **Data source** – where the data comes from.
2. **Granularity** – report-level data versus aggregated counts.
3. **Name normalization and cleaning** – handling of drug-name variants, duplicates, and incomplete reports.
4. **Signal analysis** – built-in statistical analysis of drug–event associations.
5. **Access and update frequency** – cost, restrictions, and how current the data is.

Table 2.2 applies these criteria to each system. It appears on the next page, which is laid out in landscape orientation so that the table remains readable.

2.4.2 Overview of Reviewed Systems

Table 2.2: Comparison of the reviewed systems with the proposed platform

System	Provider and data source	Granularity	Normalization and cleaning	Signal analysis	Access and updates
FAERS Public Dashboard	FDA; FAERS reports for drugs and therapeutic biologics [1]	Interactive web tool for querying FAERS data; report counts by quarter and month, search by reaction, and data export [19]	FDA states that the data contain duplicate reports and some incomplete reports [6]; reports are not extensively validated [8]	No built-in disproportionality statistics identified in the reviewed sources; FDA separately publishes quarterly Potential Signals reports [6]	Public access; FDA states that the data are updated daily (near real-time) [6]; FDA moved from quarterly to daily updates in August 2025 [18]
openFDA Drug Event API	FDA; FAERS data from 2004 onward [8]	Report-level records through an API [8]	Reports are not extensively validated [8]	Not an analysis tool	Public API; documentation notes that data may lag by three months or more depending on the quarterly FAERS release [8]
OpenVigil 2 / OpenVigil FDA	Academic open-source tool; works on FAERS data (cleaned in OpenVigil 2, openFDA-based in OpenVigil FDA) [16], [20]	Report-level data with aggregated results	OpenVigil 2 uses data cleaned with DrugBank and Drugs@FDA, and includes a drug-name mapper based on RxNorm [20], [17]	PRR, ROR, and RRR disproportionality measures [17]	Free (GPL); depends on external data sources and does not collect pharmacovigilance data itself [20], [17]
AEOLUS (Banda et al., 2016)	Academic research resource; curated version of FAERS [2]	Report-level curated data with pre-computed drug–outcome statistics [2]	Removes duplicate case records; drug names mapped to RxNorm and outcomes to SNOMED CT [2]	Pre-computed summary statistics about drug–outcome relationships [2]	Data and source code released publicly [2]; published in 2016, so current coverage should be checked
VigiBase / VigiAccess	WHO and Uppsala Monitoring Centre; global spontaneous reports from WHO PIDM members [21]	VigiAccess gives public, aggregated counts only; individual reports are not shown [15]	Standardized terminologies (WHODrug, MedDRA) and completeness checks on incoming reports [21]	VigiBase supports members in identifying safety signals [21]; VigiAccess cannot be used to compare products or to determine the likelihood of a side effect [15]	VigiAccess is public but limited; the full VigiBase dataset is subject to strict data access conditions [22]
Proposed platform	FDA FAERS quarterly data files and RxNorm (Section 3.4)	Report-level data linked to standardized drug concepts	Planned: RxNorm mapping (target of at least 90%), latest-version deduplication, and data-quality checks (Sections 3.2 and 3.3)	Planned: PRR and ROR on the standardized data, presented as potential signals	Planned: interactive dashboard with drug and reaction search, trends, and filters (Section 3.2)

The last row summarizes the platform's planned design as specified in Sections 1.4 and 3.2 to 3.5.

2.4.3 Strengths and Drawbacks of Each System

FAERS Public Dashboard. It is FDA's interactive web tool for querying FAERS data in a user-friendly way [19]. Its main drawback is that FAERS itself has limitations: FDA states that the data contain duplicate reports and that a report does not establish that a drug caused an event [6], and openFDA adds that reports are not extensively validated [8].

openFDA API. It gives programmatic access to FAERS records, which makes it a convenient ingestion source for a pipeline [8]. However, its documentation warns that reports are not extensively validated, that many drugs appear often simply because many people take them, and that FDA does not receive reports for every event [8].

OpenVigil. It is the closest existing tool to the proposed platform: it combines cleaned FAERS data, RxNorm-based name mapping, and disproportionality statistics [20], [17]. Its drawbacks are that it is an analysis tool that relies on external data sources, and that its documentation says all results should be considered unvalidated and that its installations do not use weekly updated data, so monitoring newly approved drugs is usually not readily possible [17].

AEOLUS. It is the closest published precedent for the platform's data-engineering core: a curated version of FAERS with duplicate case records removed and drug names mapped to RxNorm, released together with its source code [2]. The sources reviewed describe it as a downloadable dataset with pre-computed statistics, not as an interactive dashboard, and it was published in 2016, so its current coverage must be checked before reuse [2].

VigiBase and VigiAccess. VigiBase is described by UMC as the world's largest source of post-marketing safety data, with more than 40 million reports from over 180 WHO PIDM members as of February 2025 [21]. The public interface, VigiAccess, is limited to aggregated counts, which restricts detailed analysis [15].

2.5 Limitations and Research Gap

2.5.1 Limitations of Existing Solutions

- **Data quality is left to the user.** FDA states that its public data contain duplicate reports and some incomplete reports [6], the openFDA documentation adds that reports are not extensively validated and that updates arrive as separate reports [8], and curated resources remove duplicate case records [2].
- **Public international data is aggregated.** VigiAccess does not expose individual reports, and cannot be used to compare products or to determine the likelihood of a side effect [15].
- **Analysis tools depend on external data.** OpenVigil does not gather its own data and relies on external sources [17]. AEOLUS was published in 2016, and we did not verify whether it has been updated since [2].

2.5.2 Research Gap

Among the systems reviewed, the closest precedents are AEOLUS, an open curated FAERS dataset with duplicate removal and RxNorm mapping [2], and OpenVigil, which works on cleaned FAERS data, offers RxNorm-based name mapping, and provides disproportionality statistics [20], [17]. We did not find a reviewed system that combines (1) ingestion of FAERS reports, (2) standardization of drug names against RxNorm, (3) duplicate removal, (4) linking each report to the correct

standardized drug, (5) measured data-quality reporting, such as mapping coverage and the number of removed records, and (6) an interactive dashboard for pattern analysis, in one documented and reproducible workflow. Existing work therefore covers parts of this chain, and the proposed platform builds on it rather than claiming to be the first to normalize or deduplicate FAERS data.

Table 2.3: Gaps in existing solutions and how the proposed platform addresses them

Gap in existing solutions	How the proposed platform addresses it
FAERS data contain duplicate and incomplete reports and are not extensively validated [6], [8]	Deduplication and data-quality validation inside the pipeline
Drug names vary between reports	RxNorm-based standardization so that reports map to the correct drug [12]
Public tools offer limited analysis: VigiAccess gives aggregated counts only [15], and the FAERS dashboard is a query tool without built-in signal statistics in the reviewed sources [19]	Pattern analysis over standardized, report-level data

2.5.3 What the Project Can Learn from Existing Work

- Use disproportionality measures such as PRR and ROR, as in OpenVigil [17].
- Use RxNorm for name mapping, as OpenVigil's mapper [20] and the AEOLUS pipeline [2] do.
- Keep FAERS reactions in MedDRA terms, as FAERS does [1].
- Apply completeness checks on incoming reports, as VigiBase does [21].
- Show clear disclaimers on the dashboard, as FDA does [6].

2.5.4 Limitations of the Proposed Platform

- FAERS is a spontaneous-reporting system whose reports are unverified, so results indicate potential signals, not proven causal effects [8].
- Under-reporting is expected, because FDA does not receive a report for every adverse event [8].
- Duplicate detection can reduce but not fully eliminate duplicates.
- RxNorm covers prescription and many over-the-counter drugs available in the United States [12], so products marketed elsewhere may not map to RxNorm concepts.

3. Requirements Gathering

The Requirements Gathering phase is a critical foundation for the development of the Drug-Safety Intelligence Platform. It translates the project's objectives into detailed, actionable specifications to ensure the final system is robust, scalable, and meets the stakeholders' needs. The stakeholders who shape the requirements are identified first (Section 3.1). The requirements themselves are then categorized into six main dimensions (Sections 3.2 to 3.7): functional, non-functional, data, technical, constraints, and acceptance criteria.

3.1 Stakeholders

The needs below are the team's expected requirements, derived from the stakeholders' roles.

Table 3.1: Stakeholders and their expected needs

Stakeholder	Role	Expected needs from the platform
Pharmacovigilance specialists and regulators (for example FDA and the Egyptian Drug Authority)	Monitor post-marketing drug safety [1], [23]	Deduplicated, standardized data; signal statistics; traceable data lineage
Healthcare professionals (physicians, pharmacists)	Can report adverse events to FDA or to manufacturers [8]	Clear, searchable drug–reaction summaries with caveats
Pharmaceutical manufacturers	Normally required to send the adverse-event reports they receive to FDA [8]	Monitoring of the safety profile of their products
Researchers and academics	Use FAERS for disproportionality studies [16]	Reproducible, documented cleaned datasets and export options
Patients and consumers	Can submit reports to FDA [8] and to the Egyptian Drug Authority [23]	Understandable information with clear disclaimers
Project supervisors and evaluators	Assess the project	Complete documentation and a working demonstration
Development team (data engineers)	Build and maintain the pipeline	Maintainable pipeline, quality checks, and monitoring

In Egypt, the Egyptian Pharmacovigilance Center, affiliated with the Central Administration for Pharmaceutical Care at the Egyptian Drug Authority, assesses and follows up on reported cases [23]. The Authority's 2025 annual safety report records 51,285 adverse drug reaction cases for 2025, including 39,767 cases in which a drug was suspected [24].

3.2 Functional Requirements

Functional requirements define the core capabilities and specific behaviors the system must execute. For this platform, they include:

- **FR-01 Data Ingestion:** The system must support automated batch ingestion of FDA FAERS quarterly data files. The openFDA API is considered as an alternative route (Section 2.2.7).
- **FR-02 Data Processing & Deduplication:** The system must automatically clean the data and remove duplicate reports based on the latest case version to ensure analytical accuracy.

- **FR-03 Drug Normalization:** The system must map raw, free-text drug names to standardized RxNorm identifiers (RxCUI) using automated matching techniques.
- **FR-04 Signal Detection:** The system must calculate statistical disproportionality metrics, specifically the Proportional Reporting Ratio (PRR) and Reporting Odds Ratio (ROR), to identify potential adverse drug event signals.
- **FR-05 Data Quality Reporting:** The system must record and report data-quality metrics, including the number of records removed as duplicates and the RxNorm mapping coverage, and must keep unmapped drug names in the dataset with an explicit status flag (Section 2.2.7).
- **FR-06 Interactive Dashboard:** The system must provide drug and reaction search, top adverse reactions, drug–reaction relationships, trends over time, and filters (Section 1.4), and must display a notice that results describe reported associations and are not medical advice.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system:

- **NFR-01 Performance:** The system must be capable of processing millions of records and returning drug safety signal queries in under 3 seconds (average response time over 20 test queries, as in S3).
- **NFR-02 Scalability:** The architecture must be containerized (for example, using Docker) to allow for horizontal scaling as data volumes grow.
- **NFR-03 Data Accuracy:** The system must maintain a high drug mapping success rate (at least 90%, as in S1 and AC-03, with a target of up to 95%).
- **NFR-04 Auditability:** The system must maintain data lineage logs, ensuring that any generated safety signal can be traced back to its original FAERS record.

3.4 Data Requirements

This section outlines the data assets required for the platform's operation:

- **Raw Data:** FDA FAERS quarterly data files, which consist of seven related files (demographics, drugs, reactions, outcomes, report sources, therapy dates, and indications; see Section 2.2.2), distributed in ASCII format, with XML also available.
- **Reference Data:** Standardized terminologies and ontologies, including RxNorm (for drug normalization) and MedDRA (for adverse event coding).
- **Output Data:** Structured analytical tables containing drug_concept_id, reaction_concept_id, PRR, ROR, and case counts.

3.5 Technical Requirements

Technical requirements define the technology stack and infrastructure needed to build and deploy the system:

- **Programming Languages:** Python for ETL pipeline development and SQL for database querying.

- **Database:** PostgreSQL, chosen for its robust handling of relational data and support for JSONB and text search.
- **Tools & Libraries:** OHDSI Usagi for terminology mapping, and dbt (data build tool) for data transformation.
- **Storage:** Parquet format for efficient raw data caching and faster query performance.
- **Dashboard:** The front-end technology will be selected during the System Design phase (Section 1.5).

3.6 Project Constraints

Constraints represent the limitations and boundaries within which the project must operate:

- **Data Constraints:** FAERS data is self-reported; therefore, statistical signals indicate associations, not proven causality. The system's outputs must explicitly reflect this limitation (Section 2.2.6).
- **Licensing Constraints:** Certain commercial drug interaction APIs are unavailable or restricted; thus, drug-drug interaction checking is excluded from the current scope.
- **Resource Constraints:** The system must be deployable in a resource-limited environment without reliance on costly commercial datasets.

3.7 Acceptance Criteria

Acceptance criteria are the measurable conditions that the system must meet to be accepted by stakeholders:

Table 3.2: Acceptance criteria

ID	Criterion
AC-01	Successful automated ingestion and cleaning of one year of FAERS data with zero duplicate cases, meaning that only the latest version of each case is retained. Other suspected duplicates are removed at the rate defined in S2 (Table 1.5), because duplicate detection can reduce but not fully eliminate duplicates (Section 2.5.4).
AC-02	System ability to return relevant safety signals for a queried drug in under 3 seconds.
AC-03	A minimum of 90% drug mapping coverage on a test set of 1,000 records.
AC-04	Successful deployment and execution of the complete system via a single Docker command (docker-compose up).

References

[1] U.S. Food and Drug Administration, “FDA Adverse Event Monitoring System (AEMS) [formerly FDA Adverse Event Reporting System (FAERS)].” Accessed: Oct. 17, 2026. [Online]. Available: https://www.fda.gov/drugs/surveillance/fdas-adverse-event-reporting-system-faers

[2] J. M. Banda, L. Evans, R. S. Vanguri, N. P. Tatonetti, P. B. Ryan, and N. H. Shah, “A curated and standardized adverse drug event resource to accelerate drug safety research,” Sci. Data, vol. 3, Art. no. 160026, 2016, doi: https://doi.org/10.1038/sdata.2016.26. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC4872271

[3] World Health Organization, Global Advisory Committee on Vaccine Safety, “Vaccine safety signals from the Uppsala Monitoring Centre database.” Accessed: Oct. 17, 2026. [Online]. Available: https://www.who.int/groups/global-advisory-committee-on-vaccine-safety/topics/aefi/umc

[4] S. J. W. Evans, P. C. Waller, and S. Davis, “Use of proportional reporting ratios (PRRs) for signal generation from spontaneous adverse drug reaction reports,” Pharmacoepidemiol. Drug Saf., vol. 10, no. 6, pp. 483–486, 2001, doi: https://doi.org/10.1002/pds.677.

[5] A. Bate and S. J. W. Evans, “Quantitative signal detection using spontaneous ADR reporting,” Pharmacoepidemiol. Drug Saf., vol. 18, no. 6, pp. 427–436, 2009, doi: https://doi.org/10.1002/pds.1742.

[6] U.S. Food and Drug Administration, “AEMS Public Dashboard – FAQ [formerly FAERS Public Dashboard: Frequently asked questions].” Accessed: Oct. 17, 2026. [Online]. Available: https://fis.fda.gov/extensions/FPD-FAQ/FPD-FAQ.html

[7] U.S. Food and Drug Administration, openFDA, “FDA Adverse Event Reporting System.” Accessed: Oct. 17, 2026. [Online]. Available: https://open.fda.gov/data/faers

[8] U.S. Food and Drug Administration, openFDA, “Drug adverse event overview.” Accessed: Oct. 17, 2026. [Online]. Available: https://open.fda.gov/apis/drug/event/

[9] U.S. Food and Drug Administration, Center for Drug Evaluation and Research, “FDA Adverse Event Reporting System (FAERS): Clinical caveats and notes for users converting from LAERS to FAERS” (quarterly data extract readme, 2014 edition, as hosted by the National Bureau of Economic Research). Accessed: Oct. 17, 2026. [Online]. Available: https://m.nber.org/fda/faers/2014/Readme.pdf

[10] Uppsala Monitoring Centre, “Guideline on using VigiBase in studies” (document title derived from the published file name). Accessed: Oct. 17, 2026. [Online]. Available: https://www.who-umc.org/media/05kldqpj/guidelineusingvigibaseinstudies.pdf

[11] U.S. National Library of Medicine, “RxNorm technical documentation.” Accessed: Oct. 17, 2026. [Online]. Available: https://www.nlm.nih.gov/research/umls/rxnorm/docs/techdoc.html

[12] U.S. National Library of Medicine, “RxNorm overview.” Accessed: Oct. 17, 2026. [Online]. Available: https://www.nlm.nih.gov/research/umls/rxnorm/overview.html

[13] U.S. National Library of Medicine, “Health data standards and terminologies: A tutorial (RxNorm).” Accessed: Oct. 17, 2026. [Online]. Available: https://www.nlm.nih.gov/oet/ed/healthdatastandards/02-520.html

[14] Dryad, “Making open data useful: A drug safety case study,” Dryad blog, Jul. 12, 2016. Accessed: Oct. 17, 2026. [Online]. Available: https://blog.datadryad.org/2016/07/12/making-open-data-useful-a-drug-safety-case-study/

[15] Uppsala Monitoring Centre and World Health Organization, “About VigiAccess.” Accessed: Oct. 17, 2026. [Online]. Available: https://www.vigiaccess.org/

[16] R. Böhm, L. von Hehn, T. Herdegen, H.-J. Klein, O. Bruhn, H. Petri, and J. Höcker, “OpenVigil FDA – Inspection of U.S. American adverse drug events pharmacovigilance data and novel clinical applications,” PLoS ONE, vol. 11, no. 6, Art. no. e0157753, 2016, doi: https://doi.org/10.1371/journal.pone.0157753.

[17] “OpenVigil” (project web page and software repository). Accessed: Oct. 17, 2026. [Online]. Available: https://openvigil.sourceforge.net/

[18] Fierce Pharma, “FDA rolls out daily updates to adverse event dashboard, plans to further ‘streamline’ system,” 2025. Accessed: Oct. 17, 2026. [Online]. Available: https://www.fiercepharma.com/pharma/fda-rolls-out-daily-updates-adverse-event-dashboard-plans-further-streamline-system

[19] U.S. Food and Drug Administration, “An update to the FDA Adverse Event Reporting System (FAERS) Public Dashboard.” Accessed: Oct. 17, 2026. [Online]. Available: https://www.fda.gov/about-fda/update-fda-adverse-event-reporting-system-faers-public-dashboard

[20] “OpenVigil project page,” SourceForge. Accessed: Oct. 17, 2026. [Online]. Available: https://www.sourceforge.net/projects/openvigil

[21] Uppsala Monitoring Centre, “About VigiBase.” Accessed: Oct. 17, 2026. [Online]. Available: https://who-umc.org/vigibase-data-access/about-vigibase/

[22] Uppsala Monitoring Centre, “Frequently asked questions about VigiBase.” Accessed: Oct. 18, 2026. [Online]. Available: https://who-umc.org/vigibase/vigibase-faqs/

[23] Egyptian Drug Authority, “Reporting adverse drug reactions and medical devices.” Accessed: Oct. 17, 2026. [Online]. Available: https://edaegypt.gov.eg/en/about-us/home-page-services/home-page-services/reporting-side-effects/

[24] Egyptian Drug Authority, “Annual Safety Analysis Report (ASAR): PVGA in Egypt – 2025,” Jul. 2026. Accessed: Oct. 17, 2026. [Online]. Available: https://www.edaegypt.gov.eg/media/ucdh0u2c/asar-2025-7-2025.pdf